

Mehrdad Momeni Zadeh

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TECHNICAL SKILLS

- Languages:** Python, Java, TypeScript, JavaScript, C++, Go, SQL
- Frameworks:** Spring Boot 3, Hibernate, NestJS, Next.js, React, Django, ShadCN
- Libraries:** PyTorch, scikit-learn, TensorFlow, WandB, Hydra, Lombok, OpenCV, FFmpeg, Selenium
- DevOps & Cloud:** Docker, Kubernetes, Maven, Gradle, Jenkins, Kafka, GCP, AWS (S3), Git, JWT
- Database & Storage:** PostgreSQL, MySQL, MongoDB, Firebase, H2

WORK EXPERIENCE

- DeLTA Lab — Simon Fraser University** Burnaby, BC
Undergraduate Researcher — Scientific Machine Learning Dec 2024 – May 2025
- **Second author** on **NeurIPS 2025** paper on data-efficient neural operators.
 - Boosted PDE benchmark accuracy by **12%** and cut training cost **4×** using Google Cloud TPU-v4.
 - Improved model stability for multi-step forecasts by reducing prediction drift.
 - Collaborated with a **5-person team** to streamline training pipelines, cutting setup time to under **10 minutes**.
 - Containerised the research stack with **Docker** for reproducible deployments across labs.

TECHNICAL PROJECTS

- StagePass — Full-stack event ticketing system for university events.** [GitHub](#) 🐙
Tech: Java, Spring Boot 3, Spring Security, TypeScript, OAuth2, Lombok, PostgreSQL, React, REST API
- Deployed at **SFU's 2025 Events**, processing tickets for **100+ students**.
 - Secured organizer/attendee workflows with Spring Security and OAuth2.
 - Designed REST APIs and PostgreSQL schema for event creation, ticket sales, and validation.
- ShareScreenly — Full-stack platform for screen recording and instant video sharing.** [GitHub](#) 🐙
Tech: TypeScript, Next.js 15, Node.js, PostgreSQL, Tailwind CSS, S3, Better Auth, FFmpeg
- Adopted by **100+ SFU students** for quick demo sharing, reducing setup time from **5 min → 30 sec**.
 - Integrated AWS S3 + FFmpeg for scalable video storage, compression, and metadata handling.
 - Implemented secure role-based access with Better Auth and built a responsive Tailwind UI.
- MicroGopher — Distributed microservices platform in Go for scalable systems.** [GitHub](#) 🐙
Tech: Go, Docker, Kubernetes, PostgreSQL, MongoDB, gRPC, AMQP
- Split a monolith into modular Go microservices, improving fault isolation and performance.
 - Built services for auth, logging, and mailer communicating via REST and gRPC.
 - Deployed on Docker Swarm + Kubernetes, sustaining **10K+ requests/sec** with minimal downtime.
- HirePrep — AI-powered mock interview platform with real-time, voice-based practice.** [GitHub](#) 🐙
Tech: TypeScript, Next.js 15, Tailwind CSS, Firebase, Gemini AI, Vapi, ShadCN, Zod
- Piloted with **50+ SFU students**, improving interview performance scores by **20%**.
 - Implemented Firebase + Firestore for secure account/session management.
 - Generated tailored interview questions and feedback using Gemini AI.
- NeuroDriver — 2D driving simulator with neural agents trained via genetic algorithms.** [GitHub](#) 🐙
Tech: Python, Box2D, Genetic Algorithms, Makefile, NumPy
- Trained neural agents in a Box2D simulator, reducing collisions **20%** in **50** generations.
 - Boosted lap completion **30%** with a multi-factor reward system (speed, safety, lane discipline).
 - Automated hyper-parameter sweeps and GIF replays via one-command Makefile pipeline.

EDUCATION

- Simon Fraser University** Burnaby, BC
Bachelor of Science in Computer Science GPA: **3.4 / 4.33**
Expected Graduation Dec 2026